
Supplementary Material for “A network of biologically inspired Rectified Spectral Units (ReSUs) learns hierarchical features without error backpropagation”

1 Derivation of past-future CCA for OU processes

Optimal choices of latent variables are not always those that minimize prediction error in an L^2 sense. Indeed, from an information theoretic perspective, it is motivated to capture components of the system state that are most correlated with future dynamics [1]. Canonical correlation analysis (CCA) both accomplishes this task *and* reduces it to a spectral problem, rendering its solution simple to compute.

Consider lag vectors $\mathbf{p}_t$ and $\mathbf{f}_t$, representing past and future intervals of the system dynamics centered at time t . In one dimension, CCA seeks a direction $\mathbf{v} \in \text{span}\{\mathbf{p}_t\}$ and $\mathbf{u} \in \text{span}\{\mathbf{f}_t\}$ maximizing

$$\text{corr}(\mathbf{u}^\top \mathbf{f}_t, \mathbf{v}^\top \mathbf{p}_t).$$

That is, CCA determines an observation of the local past, $\mathbf{v}^\top \mathbf{p}_t$, that is maximally correlated with an associated observation of the future, $\mathbf{u}^\top \mathbf{f}_t$. If $\mathbf{v}^\top \mathbf{p}_t$ is large at time t , then $\mathbf{u}^\top \mathbf{f}_t$ is expected to be large in the coming moments, representing an *IF-THEN* type observation of the external world. When determining an r -dimensional compression of the system state, CCA seeks a set of r vectors

$$\Psi = \begin{bmatrix} \mathbf{v}_1^\top \\ \mathbf{v}_2^\top \\ \vdots \\ \mathbf{v}_r^\top \end{bmatrix} \in \mathbb{R}^{r \times m}$$

that maximize the correlation of $\mathbf{p}_t$ and $\mathbf{f}_t$ within their span.

This can be formulated mathematically by the following optimization problem, which is derived in [2], and which reproduces (5) in the main text:

$$\max_{\Psi \in \mathbb{R}^{r \times m}} \|\mathbb{E}[\mathbf{C}_{ff}^{-1/2} \mathbf{f}_t(\Psi \mathbf{p}_t)^\top]\|_F, \quad \text{s.t. } \Psi \mathbf{C}_{pp} \Psi^\top = \mathbf{I}_r. \quad (1)$$

Here, $\mathbf{f}_t, \mathbf{p}_t$ are the stochastic past and future lag vectors of length m , and $\mathbf{C}_{ff} = \mathbb{E}[\mathbf{f}_t \mathbf{f}_t^\top]$, $\mathbf{C}_{fp} = \mathbb{E}[\mathbf{f}_t \mathbf{p}_t^\top]$, $\mathbf{C}_{pp} = \mathbb{E}[\mathbf{p}_t \mathbf{p}_t^\top]$. We observe that

$$\begin{aligned} \mathbb{E}[\mathbf{C}_{ff}^{-1/2} \mathbf{f}_t(\Psi \mathbf{p}_t)^\top] &= \mathbf{C}_{ff}^{-1/2} \mathbb{E}[\mathbf{f}_t \mathbf{p}_t^\top] \Psi^\top \\ &= \mathbf{C}_{ff}^{-1/2} \mathbf{C}_{fp} \Psi^\top. \end{aligned} \quad (2)$$

Applying the change of variables $\tilde{\Psi} = \Psi \mathbf{C}_{pp}^{1/2}$ to (2) and to the constraint of (1), we have

$$\mathbb{E}[\mathbf{C}_{ff}^{-1/2} \mathbf{f}_t(\Psi \mathbf{p}_t)^\top] = \mathbf{C}_{ff}^{-1/2} \mathbf{C}_{fp} \mathbf{C}_{pp}^{-1/2} \tilde{\Psi}^\top \quad \text{and} \quad \Psi \mathbf{C}_{pp} \Psi^\top = \tilde{\Psi} \tilde{\Psi}^\top. \quad (3)$$

Substituting (3) into (1) we find that the solution to optimization (1) is equal to

$$\max_{\tilde{\Psi} \in \mathbb{R}^{r \times m}} \|\mathbb{E}[\mathbf{C}_{ff}^{-1/2} \mathbf{C}_{fp} \mathbf{C}_{pp}^{-1/2} \tilde{\Psi}^\top]\|_F, \quad \text{s.t. } \tilde{\Psi} \tilde{\Psi}^\top = \mathbf{I}_r. \quad (4)$$

The solution to the reformulated optimization problem (4) is obtained by the singular value decomposition (SVD) $\mathbf{C}_{ff}^{-1/2} \mathbf{C}_{fp} \mathbf{C}_{pp}^{-1/2} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top$ [3]. Assuming the singular values on the diagonal of $\mathbf{\Sigma}$ are ordered from largest to smallest, the optimum of (4) is achieved by the Frobenius norm of the upper-left $r \times r$ submatrix of $\mathbf{\Sigma}$. The $\tilde{\mathbf{\Psi}}$ that achieves that maximum is the matrix of the first r columns of $\mathbf{V}$, which we denote $\mathbf{V}_r$.

Furthermore, since $\mathbf{V}_r$ achieves the maximum of (4), using the change of variables $\tilde{\mathbf{\Psi}} = \mathbf{\Psi} \mathbf{C}_{pp}^{1/2}$, the maximum of (1) is achieved by $\mathbf{\Psi} = \mathbf{V}^\top \mathbf{C}_{pp}^{-1/2}$.

Derivation of mutual information between $\mathbf{z}_t$ and $\mathbf{f}$

In this section we prove that the mutual information between $\mathbf{z}_t$ and $\mathbf{f}_t$ is equation (8) in the main text:

$$I_r = -\frac{1}{2} \sum_{i=1}^r \log(1 - \sigma_i^2). \quad (5)$$

We start by observing that the covariance matrix of $[\mathbf{z}_t \ \mathbf{f}_t]$ is the $(r + m) \times (r + m)$ matrix

$$\mathbf{\Sigma}_{\mathbf{z}_t, \mathbf{f}_t} = \begin{bmatrix} \mathbf{I}_r & \mathbf{\Sigma}^\top \mathbf{U}^\top \mathbf{C}_{ff}^{1/2} \\ \mathbf{C}_{ff}^{1/2} \mathbf{U} \mathbf{\Sigma} & \mathbf{C}_{ff} \end{bmatrix}. \quad (6)$$

The off-diagonal blocks are obtained via the equations

$$\mathbb{E}[\mathbf{f}_t \mathbf{z}_t^\top] = \mathbb{E}[\mathbf{f}_t (\mathbf{\Psi} \mathbf{p}_t)^\top] \quad (7)$$

$$= \mathbb{E}[\mathbf{f}_t \mathbf{p}_t^\top] \mathbf{\Psi}^\top. \quad (8)$$

Using the definition of $\mathbf{\Psi}$ and that $\mathbf{C}_{ff}^{-1/2} \mathbf{C}_{fp} \mathbf{C}_{pp}^{-1/2} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top$, we have

$$\begin{aligned} \mathbb{E}[\mathbf{f}_t \mathbf{z}_t^\top] &= \mathbf{C}_{fp} \mathbf{C}_{pp}^{-1/2} \mathbf{V} \\ &= \mathbf{C}_{ff}^{1/2} \mathbf{C}_{ff}^{-1/2} \mathbf{C}_{fp} \mathbf{C}_{pp}^{-1/2} \mathbf{V} \\ &= \mathbf{C}_{ff}^{1/2} \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top \mathbf{V} \\ &= \mathbf{C}_{ff}^{1/2} \mathbf{U} \mathbf{\Sigma}. \end{aligned} \quad (9)$$

Using the formula for the mutual information between Gaussian vectors [4], we obtain

$$I_r = \frac{1}{2} \log \left(\frac{|\mathbf{I}_r| |\mathbf{C}_{ff}|}{|\mathbf{\Sigma}_{\mathbf{z}_t, \mathbf{f}_t}|} \right) \quad (10)$$

where $|\cdot|$ denotes the matrix determinant. Substituting the formula for the determinant of a 2×2 block matrix [5], we have

$$|\mathbf{\Sigma}_{\mathbf{z}_t, \mathbf{f}_t}| = |\mathbf{I}_r| |\mathbf{C}_{ff} - (\mathbf{C}_{ff}^{1/2} \mathbf{U} \mathbf{\Sigma})(\mathbf{\Sigma}^\top \mathbf{U}^\top \mathbf{C}_{ff}^{1/2})|. \quad (11)$$

Applying the determinant property $|AB| = |A||B|$ along with the fact that orthogonal matrices have determinant of one, we have

$$\begin{aligned} |\mathbf{\Sigma}_{\mathbf{z}_t, \mathbf{f}_t}| &= |\mathbf{C}_{ff}^{1/2} (\mathbf{I}_r - \mathbf{U} \mathbf{\Sigma} \mathbf{\Sigma}^\top \mathbf{U}^\top) \mathbf{C}_{ff}^{1/2}| \\ &= |\mathbf{C}_{ff}| |(\mathbf{I}_r - \mathbf{U} \mathbf{\Sigma} \mathbf{\Sigma}^\top \mathbf{U}^\top)| \\ &= |\mathbf{C}_{ff}| |\mathbf{U} (\mathbf{I}_r - \mathbf{\Sigma} \mathbf{\Sigma}^\top) \mathbf{U}^\top| \\ &= |\mathbf{C}_{ff}| |(\mathbf{I}_r - \mathbf{\Sigma} \mathbf{\Sigma}^\top)|. \end{aligned} \quad (12)$$

Applying equation (12) to (10), we obtain

$$I_r = \frac{1}{2} \log \left(\frac{1}{|\mathbf{I}_r - \mathbf{\Sigma} \mathbf{\Sigma}^\top|} \right) = -\frac{1}{2} \sum_{i=1}^r \log(1 - \sigma_i^2) \quad (13)$$

where $\sigma_i = \Sigma_{i,i}$.

2 Gaussian processes

In Section 2 of the main text, we observe that CCA can be extended to time-invariant Gaussian processes (GPs) with integrable covariance kernel as long as the past vector is sufficiently long. Here, we numerically test the necessary length ℓ of the past lag vector $\mathbf{p}_t$ that incorporates enough information that CCA is approximately optimal in the sense of equation (5).

We start by constructing a grid $t_j = \Delta_t j$ with $\Delta_t = 0.05$ for $i = 1, \dots, 2\ell$.¹ We then we construct the $2\ell \times 2\ell$ GP covariance matrix

$$\Sigma = \begin{bmatrix} \mathbf{C}_{pp} & \mathbf{C}_{pf} \\ \mathbf{C}_{fp} & \mathbf{C}_{ff} \end{bmatrix} \quad (14)$$

where $\Sigma_{ij} = k(t_i - t_j)$. Here, k is with rational quadratic covariance kernel

$$k(t - t') = \left(1 + \frac{|t - t'|^2}{2\alpha l^2}\right)^{-\alpha} \quad (15)$$

with $l = \alpha = 1$ used in the main text. We compute the canonical directions Ψ and the canonical correlations $\sigma_1, \dots, \sigma_\ell$ as described in Section 1 of the main text. We show that for sufficiently large ℓ , increasing ℓ adds negligible information about the future. That is, for large ℓ , increasing ℓ further results in adding near-zero components to canonical directions Ψ and also results in adding zero canonical correlation coefficients. Specifically, we compare the canonical correlations and directions for lag ℓ and compare them to those computed for lag 2ℓ . Throughout this section, we set the past and future lags to both be of the same size, though fixing the size of the future lag vector does not change the results.

In Fig. S1A, we compare $\ell = 5$ and $\ell = 10$. Even a lag of $\ell = 5$ captures significant information about the future, but there is substantial information lost compared to $\ell = 10$ (see the difference between the fourth canonical correlations). We compare $\ell = 20$ and $\ell = 40$ in Fig. S1B. With a lag of $\ell = 20$, CCA is well-approximated by the CCA for $\ell = 40$. Furthermore, the number of canonical correlations needed for capturing nearly all information about the future is contained in five canonical directions. We plot convergence of the first canonical direction Ψ_1 . The convergence of the next canonical directions is similar to that of the first canonical direction.

We also compute the mutual information of past and future lag vectors $\mathbf{p}_t, \mathbf{f}_t$ for various ℓ using equation (8). The results in Fig. S1C demonstrate that the mutual information rapidly approaches a limit. This reinforces that a lag of around $\ell = 20$ captures nearly all information about the future.

The results of this section, unlike in the main text, compute CCA with access to the exact covariance matrix in contrast to a covariance matrix approximated from data. We observe numerically that with more observation noise, there are fewer canonical correlations near one. Intuitively, the more noisy the past and future observations are, the less correlated the past will be with the future.

For many commonly-used GP covariance kernels (e.g., squared exponential, rational quadratic, Matérn, etc.), the cross-covariance matrix $\mathbf{C}_{fp}$ is numerically low-rank. That is, the singular values of $\mathbf{C}_{fp}$ decay quickly. This phenomenon is a result of the fact that covariance kernels tend to be smooth (in the sense of frequency content) away from zero and is discussed at length in [6]. The matrix $\mathbf{C}_{fp}$ being numerically low-rank does not guarantee that the matrix $\mathbf{C}_{ff}^{-1/2} \mathbf{C}_{fp} \mathbf{C}_{pp}^{-1/2}$ is also numerically low-rank. However, we observe empirically that this tends to be the case and equivalently that past/future CCA for GPs tends to produce few canonical correlations near one. We demonstrate this property of the rational quadratic kernel in the second column of Fig S1.

3 Details of numerical simulations

3.1 Contrast from natural scenes

We used a natural scene dataset from [7], which can be downloaded from here. This dataset consists of 421 panoramic luminance-calibrated naturalistic images of natural scenes. Each image contains 927×251 pixels and subtends 360° horizontally and 97.4° vertically.

¹The conclusions of this section do not depend on the choice of Δ_t . The lag vector lengths discussed here are only relevant for a particular choice of Δ_t . If Δ_t is multiplied by a constant, conclusions hold when dividing the lag by the same constant.

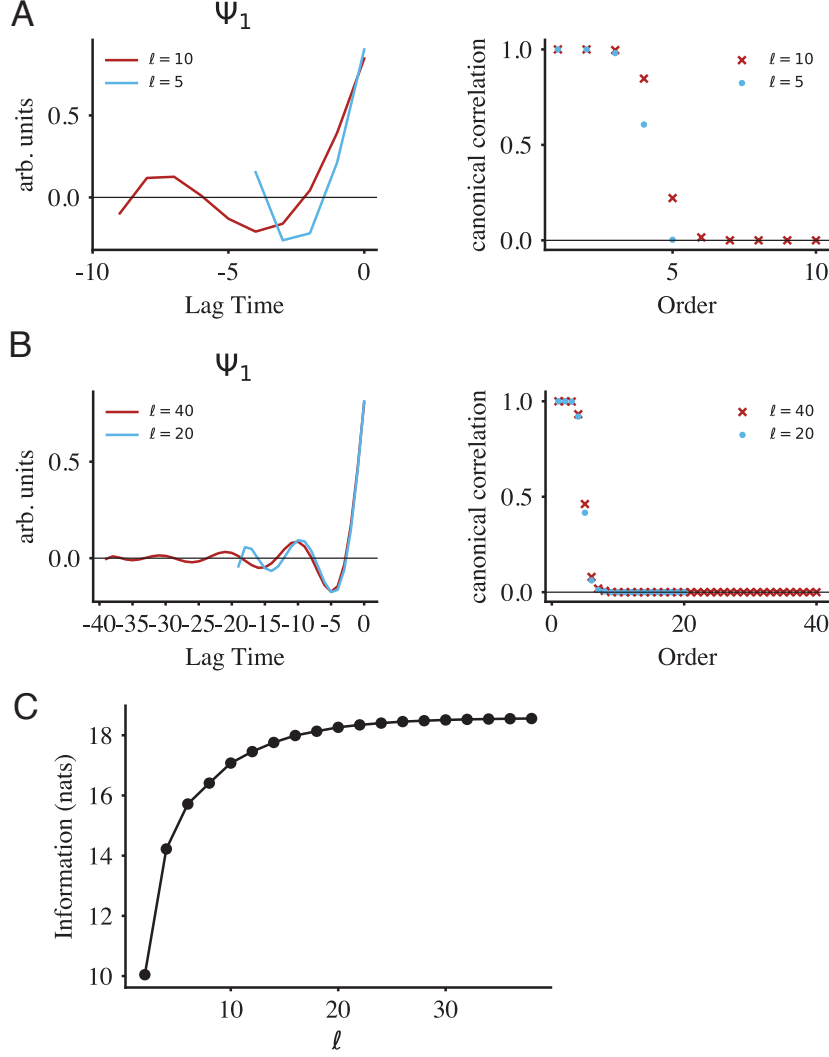


Figure S1: CCA for a GP with rational quadratic kernel converges as the lag length, ℓ , increases. We demonstrate the convergence of temporal filters (or canonical directions) Ψ_1 in the top row and in the second row we plot the corresponding correlation coefficients. (A) For $\ell = 5$ and $\ell = 10$, there is still substantial disagreement in the fourth and fifth canonical directions and in Ψ_1 . (B) For $\ell = 20$, the filter Ψ_1 and the canonical directions already closely approximate the $\ell = 40$ case. (C) The mutual information between p_t and f_t converges as we increase the lag length ℓ .

To simulate the optical system and photoreceptor transduction of *Drosophila*, the grayscale luminance images were converted into contrast images with a blurring step and contrast computation step as outlined in [8, 9]. Briefly, the pixel values of original pictures, denoted as I , were blurred with a two-dimensional Gaussian filter $f_{\text{blur}}(x, y)$

$$I_{\text{blur}} = I * f_{\text{blur}}(x, y), \quad (16)$$

$$f_{\text{blur}}(u, v) = \frac{1}{2\pi\lambda_{\text{blur}}^2} \exp\left(-\frac{u^2 + v^2}{2\lambda_{\text{blur}}^2}\right), \quad u_- \leq u \leq u_+, \quad v_- \leq v \leq v_+. \quad (17)$$

The filter extends to $\pm 3\lambda_{\text{blur}}$, i.e., $u_+ = v_+ = |u_-| = |v_-|$, where $\lambda_{\text{blur}} = \frac{5.3}{2\sqrt{2 \ln 2}}$ is chosen to mimic the spatial resolution of the fly's ommatidia, and $*$ denotes convolution.

Then, the blurred images were converted to contrast signals

$$c(x, y) = \frac{I_{\text{blur}} - I_{\text{mean}}(x, y)}{I_{\text{mean}}(x, y)}, \quad (18)$$

$$I_{\text{mean}}(x, y) = I_{\text{blur}} * f_{\text{local-mean}} \quad (19)$$

where $f_{\text{local-mean}}$ is a 2-dimensional Gaussian filter similar to that of f_{blur} but with a larger spatial scale than λ_{blur} [8]. Scalar contrast time series was generated by scanning a selected row of the contrast image with constant velocity. A Matlab code convert raw pixel pictures to blurred contrast is available at: <https://github.com/ClarkLabCode/SynapticModel/tree/master>.

3.2 Comparing the response to staircase stimuli

In fly vision study, staircase luminance/contrast stimuli are widely used to probe the temporal properties of early visual neurons. Since our model accounts for the response of neurons a synapse behind photoreceptors, the L1/L2/L3 neurons, the luminance was logarithmically transformed to approximate the responses of photoreceptors [10].

The staircase stimulus was generated by concatenating up and down transients. Each upward transient is generated by hyperbolic function $\tanh(x)/2 + b_0$ with x uniformly taking value from -10 to 10 and the constant base b_0 runs from 1 to 3 indicating the elevated plateaus. The downward transients are mirror of the upward transients. A small observation noise was also introduced ($\sigma = 0.01$), the second column of figure 4 shows the average predicted responses from our model.

3.3 Training a two-layer ReSU network on natural scene

To demonstrate that a two layer ResU network can perform non-trivial computation. We trained the network with contrast profiles from natural scene. The input to the first layer ReSU are lag vectors $\mathbf{p}_t, \mathbf{f}_t$ of scalar image contrast x_t as described in the previous subsection. In our numerical simulation, both the memory and horizon are set to be 50 and the observation noise level is 0.05. The output is a 2-D vector time series $\mathbf{z}_t$, the first component is the projection of input lag vector $\mathbf{p}_t$ on the first filter Ψ_1 , and the second component is the projection onto the second filter with rectification, i.e. $z_2(t) = [\Psi_2 \mathbf{p}_t]_+$. The input to the second layer network consist of output of three adjacent “channels” of first layer $\mathbf{y}_t = [z_1^L(t), z_2^L(t), z_1^C(t), z_2^C(t), z_1^R(t), z_2^R(t)]^\top \in \mathbb{R}^6$, where the superscripts L, C, R stand for left, center and right channel, respectively. This set up mimic the spatial pooling of downstream motion sensitive neurons such as T4 and T5. In practice, these three channels are separated by about 5 degree which corresponds 13 horizontal pixel on the panoramic pictures. Thus, the time series of $z_{1,2}^C(t), z_{1,2}^R(t)$ are time-shifted versions of $z_{1,2}^L(t)$. We also normalize each channel by its standard deviation and introduced small observation noise. Finally, we obtain the past and future lag vectors for the second layer $\mathbf{p}_t^2 = \mathbf{y}_t, \mathbf{f}_t^2 = \mathbf{y}_{t+\text{lag}}$, the covariance matrices are defined as $\mathbf{C}_{fp} = \mathbb{E}(\mathbf{f}_t^2 \mathbf{p}_t^{2\top})$, $\mathbf{C}_{ff} = \mathbb{E}(\mathbf{f}_t^2 \mathbf{f}_t^{2\top})$, $\mathbf{C}_{pp} = \mathbb{E}(\mathbf{p}_t^2 \mathbf{p}_t^{2\top})$, and the objective function of for the second layer is the same as Equation 6 in the main text. For the numerical results shown in the figure 5, the lag length is set to be 5.

After applying CCA on the lag vectors of the second layer, we obtained two filters for the second layer. The output of the network is then projection of $\mathbf{p}_t$ onto the second filter Ψ_2^2 . The weights of Ψ_2^2 are then compared with the synaptic weights between LMCs and T4 neurons in fly as shown in figure 5 in the main text [11].

3.4 Adaptation of filters to change of input statistics

We demonstrate that filters in the ReSU network adapt rapidly to changes in input statistics. We used time series generated from a Gaussian process with a rational quadratic kernel (equation (15)) where $\alpha = \ell = 1$. The time series had two distinct regimes: the first half had high signal-to-noise ratio (SNR, $\sigma = 0.01$) and the second half had low SNR ($\sigma = 0.1$), as shown in Figure S2A.

To quantify the adaptation speed when input statistics changed at $t = 2000$, we tracked how the “current filter” correlated with two reference filters: one estimated from the first half of the trajectory (high SNR) and one from the second half (low SNR). We estimated the current filter at each time point by computing a past-future covariance matrix from preceding input data, where data points were weighted by an exponential decay function $e^{-|t'-t|/\tau}$. This weighting scheme gives greater

importance to recent observations, with $\tau = 400$ in our simulations, where t is the current time point and t' represents previous time points.

When the input SNR changed, the filter's correlation with the first reference filter decreased while its correlation with the second reference filter increased rapidly. The filter adapted to the low-SNR statistics in approximately 300 time steps, less than 10 times the memory lag (Figure S2C).

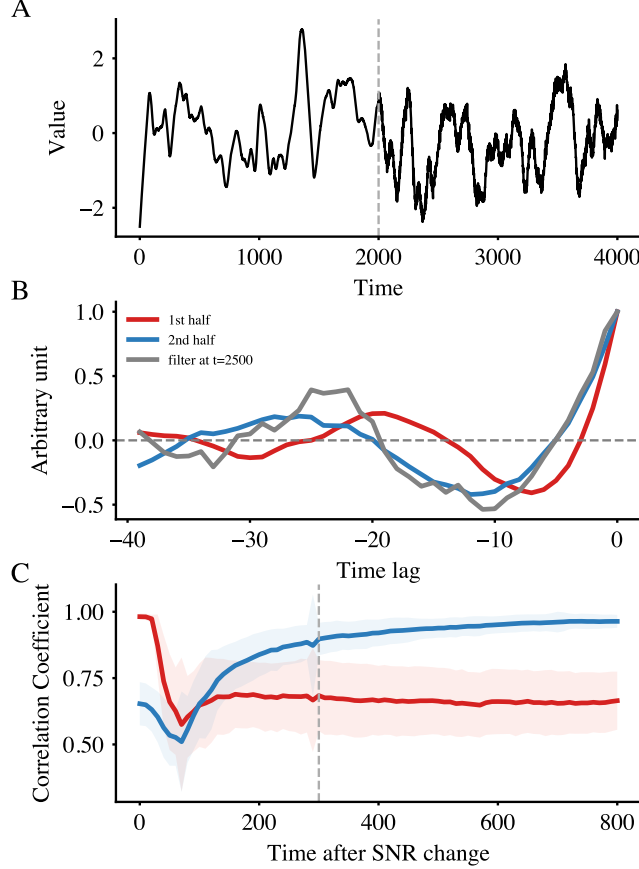


Figure S2: Filter in ReSU adapts to the input statistics quickly. (A) A input generated from GP with a rational quadratic kernel. It changes noise level at $t = 2000$. (B) The second filter obtained from offline CCA for the first half (red) and the second half (blue) trajectory. The filter at $t = 2500$ is estimated using all the data up to $t = 2500$ but with an exponential discount. (C) Correlation coefficient of the “current” filter with that of the reference filters (as shown in B). Shaded region indicates standard deviation from 100 independent simulations. Parameters: $m = 40, h = 20$.

3.5 Example of the OU process

The example of OU process shown in Figure 2 was simulated by the following system

$$X_{t+1} = AX_t + B\xi_t, \quad (20)$$

$$y_t = CX_t + \eta_t, \quad (21)$$

where

$$A = \begin{bmatrix} 0.6 & 0.6 & 0 \\ -0.6 & 0.6 & 0 \\ 0 & 0 & 0.4 \end{bmatrix}, \quad B = [0.17, -0.15, 0.28], \quad C = [0.78, 0.53, 1] \quad (22)$$

and $\xi_t \sim \mathcal{N}(0, 1), \eta_t \sim \mathcal{N}(0, 0.1)$.

3.6 Experimental data

The temporal filters of retinal ganglion cells shown in figure 3 were obtained from [12] and the dataset can be downloaded here. The synaptic weights of the ON motion detection pathway shown in figure 5 was obtained from figure 6 of [11].

3.7 Hyperparameters

We list all the hyper parameters used to learn the temporal filters in the simulations, such as the past and future lag, m , h , and the observation noise level, σ in table 1.

Table 1: parameters								
parameter	Fig. 2			Fig. 3	Fig. 4	Fig. 5	Fig. S1	Fig. S2
	B	C	D					
m	25	75	50	75	50	50		40
h	25	25	50	25	50	50		20
σ	0	0.1	0.05	0.01, 0.4	0.5, 0.05	0.05, 0.01	0	0.01, 0.1

In Fig. 4, when testing the action of filters on the staircase input, a small observation noise $\sigma = 0.002$ was added. Similarly, in Fig. 5B, a small observation noise ($\sigma = 0.005$) was added to the step-like contrast.

3.8 Code availability

All the code in this project are available at a GitHub repository: <https://github.com/ShawnQin/ReSU.git>

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